Problem 4 | Problems T6andT7_Electric and Magnetic Properties.pdf #4 | Metal vs intrinsic semiconductor · temperature

Effect of Temperature on a Metal and on an Intrinsic Semiconductor

Original statement · Problems T6andT7_Electric and Magnetic Properties.pdf, p. 2

> *4. Justify the different effects that an increase on the exposure temperature have on a metal and on an intrinsic semiconductor*
>
> *The sheet prints no solution for this problem.*

HYPOTHESES & PARAMETERS

Hypotheses

1. **Conductivity is carriers times mobility.** For any conductor $\sigma = n|q|\mu$; for an intrinsic semiconductor both electrons and holes count, $\sigma = n|q|(\mu_e + \mu_h)$ [Session 13 Slides 12, 27-28].
2. **Metal.** The conduction electrons are the non-bonded electrons of the atoms, so their number does not depend on temperature in the range considered; only the mobility changes, through the thermal resistivity $\rho_T = \rho_0(1 + \alpha_T T)$ [Session 13 Slides 13-14].
3. **Intrinsic semiconductor.** The carriers are created by thermal excitation across the gap, so $n = p = n_i$ grows with the temperature as $n_i \propto \exp(-E_g/2kT)$ and the exponential factor is the one that dominates the conductivity [Session 13 Slide 29]. The temperature also disorders the lattice, but that effect "is masked by the increase of the number of charge carriers" (Slide 29).
4. **Numerical illustration.** Copper with $\alpha_T = 0.0039$ °C⁻¹ [Session 13 Slide 14] and germanium with $E_g = 0.67$ eV [Session 13 Slide 8], both from 25 °C to 75 °C, the interval of Problem T6-07. $k_B = 8.62 \times 10^{-5}$ eV/K.

Data

Quantity	Symbol	Value
Temperature coefficient of Cu	α_T	0.0039 °C⁻¹
Band gap of Ge	E_g	0.67 eV
Boltzmann constant	k_B	8.62×10^{-5} eV/K
Temperatures	T_1, T_2	25 °C, 75 °C (298.15 K, 348.15 K)

Unknown: the sign and the size of the change of σ in each material.

MODEL & STRATEGY

Take the logarithmic derivative of $\sigma = n\mu$ with respect to the temperature: it splits the change of the conductivity into the change of the carrier number and the change of the mobility. In the metal only the second term exists; in the intrinsic semiconductor the first term is far larger than the second.

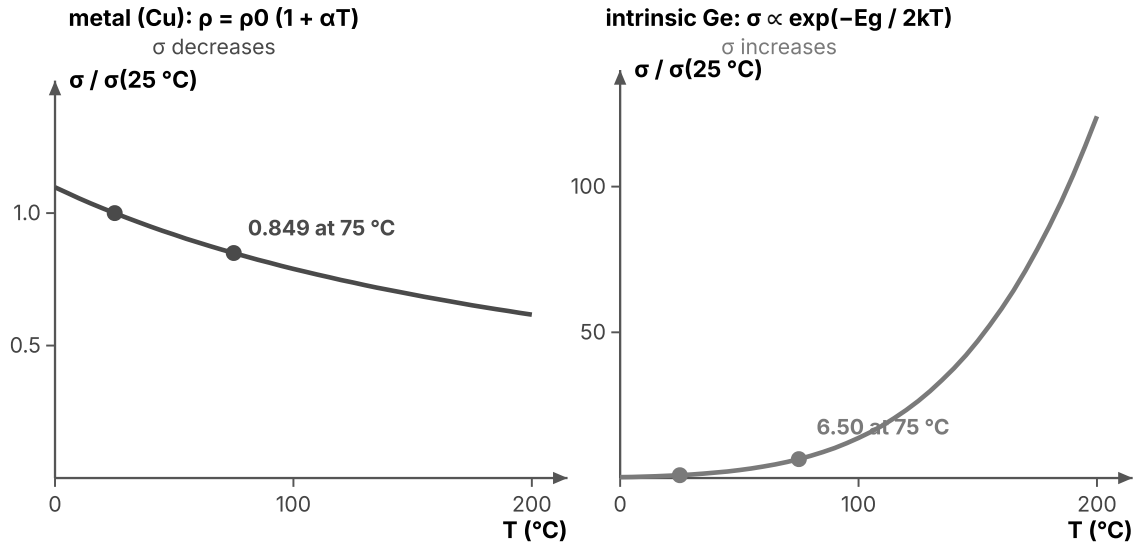


Figure 4. Problem 4: relative conductivity of copper (ρ = ρ0(1 + αT), α = 0.0039 °C^{-1}) and of intrinsic germanium (Eg = 0.67 eV) against temperature, each divided by its value at 25 °C.

Strategy.

1. Write the logarithmic derivative of the conductivity.
2. Evaluate it for the metal, using Slide 14, and for the semiconductor, using Slide 29.
3. Compare the numerical changes between 25 and 75 °C.

STEP-BY-STEP CALCULATION

1. The two ways in which temperature can act

Conductivity as number of carriers times mobility, and its logarithmic derivative (product rule of the logarithm and chain rule):

$$\sigma = n\,|q|\,\mu \;\Rightarrow\; \ln\sigma = \ln n + \ln|q| + \ln\mu$$

$$\frac{1}{\sigma}\frac{d\sigma}{dT} = \frac{1}{n}\frac{dn}{dT} + \frac{1}{\mu}\frac{d\mu}{dT}$$

2. Metal [Session 13 Slides 13-14]

The number of free electrons is constant, so the first term vanishes, and the conductivity is the inverse of the thermal resistivity:

$$\frac{dn}{dT} = 0 \qquad \sigma = \frac{1}{\rho_0 (1 + \alpha_T T)}$$

Derivative of the inverse (chain rule with $u = 1 + \alpha_T T$, $du/dT = \alpha_T$):

$$\frac{1}{\sigma}\frac{d\sigma}{dT} = -\frac{1}{u}\frac{du}{dT} = -\frac{\alpha_T}{1 + \alpha_T T}$$

Evaluated for copper at 25 °C:

$$1 + \alpha_T T = 1 + 0.0039\ {}^\circ\mathrm{C}^{-1} \times 25\ {}^\circ\mathrm{C} = 1.0975$$

$$\frac{1}{\sigma}\frac{d\sigma}{dT} = -\frac{0.0039\ {}^\circ\mathrm{C}^{-1}}{1.0975} = \mathbf{-3.554 \times 10^{-3}\ K^{-1}}$$

so the relative change of the conductivity per kelvin, written as a percentage, is:

$$-3.554 \times 10^{-3}\ \mathrm{K}^{-1} \times 100\ \% = -0.3554\ \%\ \mathrm{K}^{-1}$$

Over the 50 K of the interval:

$$\frac{\rho(75\ {}^\circ\mathrm{C})}{\rho(25\ {}^\circ\mathrm{C})} = \frac{1 + 0.0039 \times 75}{1 + 0.0039 \times 25} = \frac{1.2925}{1.0975} = 1.1777$$

$$\frac{\sigma(75\,^\circ\mathrm{C})}{\sigma(25\,^\circ\mathrm{C})} = \frac{1}{1.1777} = \mathbf{0.8491}$$

3. Intrinsic semiconductor [Session 13 Slide 29]

The conductivity follows the carrier number:

$$\sigma = \sigma_0 \exp\!\left(-\frac{E_g}{2k_B T}\right) \;\Rightarrow\; \ln\sigma = \ln\sigma_0 - \frac{E_g}{2k_B}\,T^{-1}$$

Derivative with respect to T (power rule on T^{-1}):

$$\frac{1}{\sigma}\frac{d\sigma}{dT} = -\frac{E_g}{2k_B}\left(-T^{-2}\right) = \frac{E_g}{2k_B\,T^2}$$

Evaluated for germanium at 298.15 K, first the constant $E_g/2k_B$ and the square of the temperature:

$$\frac{E_g}{2k_B} = \frac{0.67\ \mathrm{eV}}{2 \times 8.62\times10^{-5}\ \mathrm{eV/K}} = 3886.3\ \mathrm{K}$$

$$T^2 = (298.15\ \mathrm{K})^2 = 8.8893\times10^{4}\ \mathrm{K}^2$$

$$\frac{1}{\sigma}\frac{d\sigma}{dT} = \frac{3886.3\ \mathrm{K}}{8.8893\times10^{4}\ \mathrm{K}^2} = +\mathbf{4.372\times10^{-2}\ K^{-1}}$$

so the relative change of the conductivity per kelvin, written as a percentage, is:

$$4.372\times10^{-2}\ \mathrm{K^{-1}} \times 100\ \% = +4.372\ \%\ \mathrm{K^{-1}}$$

Over the 50 K of the interval (the ratio of Problem T6-07):

$$\frac{\sigma(75\ ^\circ\mathrm{C})}{\sigma(25\ ^\circ\mathrm{C})} = \exp\left[3886.3\ \mathrm{K}\left(\frac{1}{298.15\ \mathrm{K}} - \frac{1}{348.15\ \mathrm{K}}\right)\right] = \exp(1.8720) = \mathbf{6.501}$$

4. Comparison of the two

Ratio of the relative rates at 25 °C, in absolute value:

$$\frac{4.372 \times 10^{-2}\ \mathrm{K}^{-1}}{3.554 \times 10^{-3}\ \mathrm{K}^{-1}} = 12.30$$

RESULT & CHECKS

Checks

- **Units of both logarithmic rates.**

$$\left[\frac{\alpha_T}{1+\alpha_T T}\right] = \frac{^\circ\mathrm{C}^{-1}}{1} = \mathrm{K}^{-1} \qquad \left[\frac{E_g}{2k_B T^2}\right] = \frac{\mathrm{eV}}{\mathrm{eV\,K^{-1}\,K^2}} = \mathrm{K}^{-1}$$

- **Interval.** A temperature difference of 1 °C equals 1 K, so the coefficient in °C⁻¹ can be read in K⁻¹.
- **Signs.** Metal: negative rate; semiconductor: positive rate.

Interpretation

In the metal the carriers are already all there; heat only agitates the ions, the electrons collide more often, the mobility falls and so does σ. In the intrinsic semiconductor the heat itself creates carriers by lifting electrons across the gap, and the exponential growth of their number (4.4 % per kelvin) overpowers the loss of mobility, which is the same kind of effect as in the metal (a loss of 0.36 % per kelvin for copper). Over 50 K the metal loses 15 % of its conductivity and the semiconductor multiplies it by 6.5. For $E_g \to 0$ the semiconductor rate tends to zero, as in a metal where the carrier number is temperature independent; for a large gap the rate becomes very large, and the material is an insulator at ordinary temperatures.

Check against the official solution

Item	Official	Computed	Comment
Effect on the metal	not printed on the sheet	σ decreases (0.8491 for Cu, 25 to 75 °C)	consistent with Session 13 Slides 13-14

Item	Official	Computed	Comment
Effect on the intrinsic semiconductor	not printed on the sheet	σ increases (6.501 for Ge, 25 to 75 °C)	consistent with Session 13 Slide 29; the 6.501 is the ratio of Problem T6-07